University of Colombo School of Computing

Internal Undergraduate Degrees

Course Detail Document

Course Outline									
Course Code :	**IS1201**								
Course Name :	**Programming and Problem Solving**								
(Bold and underline the appropriate)	**Year :**	**1**	2	3	4		**Semester :**	**1**	2
Number of Credits	3								
Core/Optional	Core								
Evaluation Criteria	**Assignments :**	30%							
	Final Exam :	70 %							
Requisites for following the Course									
Method of Delivery						**Per Week**		**Total**	
	Lectures					2 Hours		30 Hours	
	Tutorials					1 Hour		15 Hours	
	Lab Work					2 Hour		30 Hours	
	Group Work					-		-	

Course Definition	
Introduction	This course is designed to introduce students to the fundamental concepts of programming and problem-solving techniques. It provides a strong foundation in the C programming language, enabling students to develop structured solutions to computational problems. The course emphasizes algorithmic thinking, logical reasoning, and hands-on practice to build programming proficiency.
Course Aim/Intended Learning Outcomes	Upon completion of this course, students will be able to do the following: LO1. Understand the overview and history of programming languages and different programme design and development approaches. LO2. Understand and apply techniques for modeling programme structures, algorithms, flow charts and pseudo codes.

	LO3. Understand how to use constants, variables, literals, primitive data types, and expressions. LO4. Learn how to use operators and operations, decision logic selection, simple selection and multiple selection. LO5. Learn how to use functions, parameters and passing parameters. LO6. Learn how to write codes when solving simple problems. LO7. Understand and apply control structures and recursion LO8. Understand and apply simple data structures and file handling. L09. Implement and debug applications to solve simple real life problems.
References/Reading Materials	• The C programming language, Kernighan and Brian W., ISBN 0-13-110370-9 • C Programming: A Modern Approach, K. N. King, ISBN-13: 978-0393979503 • Learn C Programming - The Definitive Guide, online available at https://www.programiz.com/c-programming • C Tutorial, online available at https://www.tutorialspoint.com/cprogramming/

Course Content

Topic 1: Overview and history of programming languages

- Computer and programming history
- Evolution of Programming Languages
- Processing a High-Level Language Program
- Introduction to Scratch
- Introduction to C Programming

Duration: 1 week

Topic 2: Programming Fundamentals

- Input/outputs
- Variables
- Constant
- Literals
- Data types
- Operators and Operations
- Expressions.

Duration: 3 weeks

Topic 3: Control structure • Simple Selection • Multiple Selection • Looping • Break and Continue **Duration: 2 weeks**
Topic 5: Functions • Macros & Pre-processer • Functions • Parameters • Passing Parameters • recursion **Duration: 3 weeks**
Topic 6: Handling Simple Data Structures • Pointers • Arrays and Strings • Structures • Unions • Enumerations **Duration: 4 weeks**
Topic 7: File Handling • files • reading • writing **Duration: 2 weeks**